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NATIONAL TSING HUA UNIVERSITY

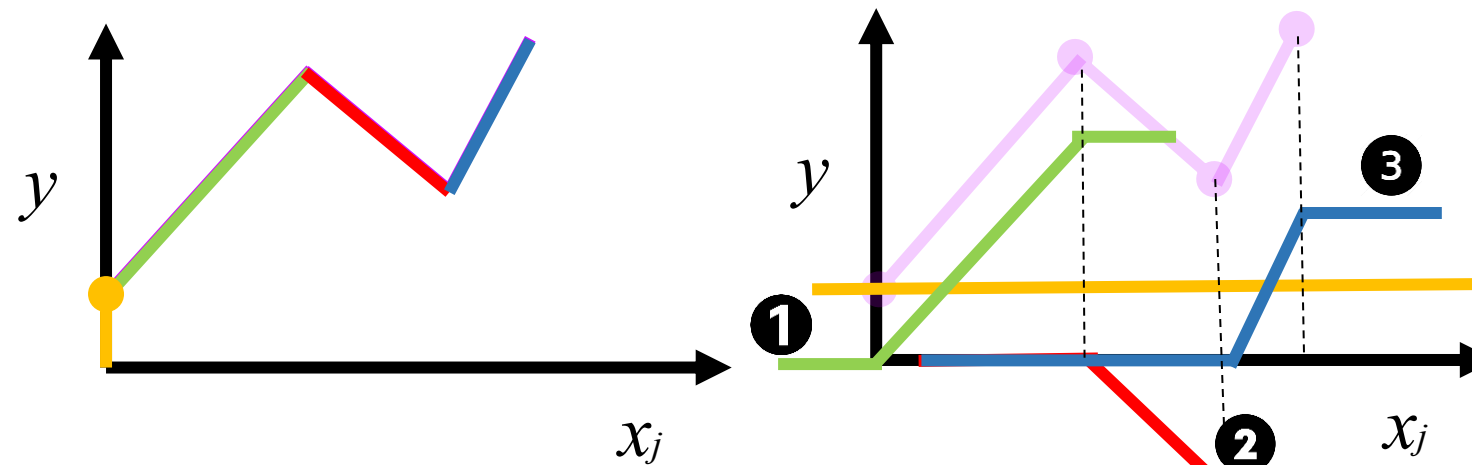
Deep Learning and Applications

Ching-I Huang



Activation Functions

- To the output of a linear transformation yields a nonlinear transformation, activation function g is typically chosen to be a function that is applied element-wise.



$$y = \text{yellow} + \text{green} + \text{red} + \text{blue}$$

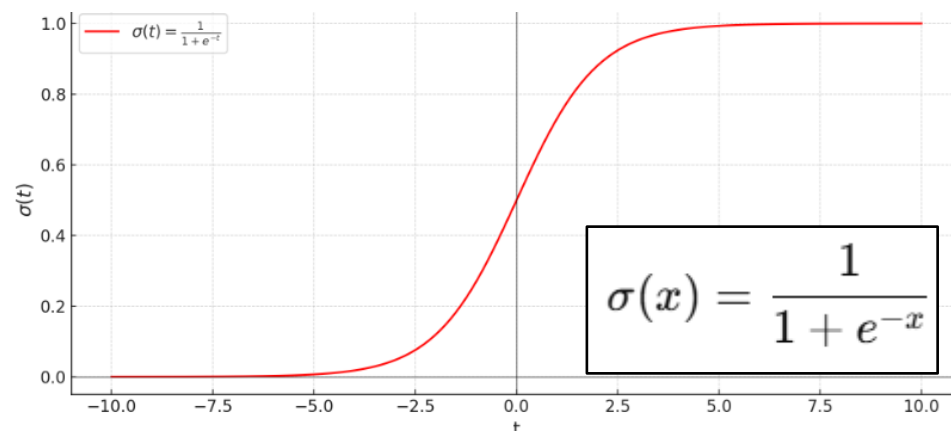
$$= b + \sum \text{segment}$$

$$y = b + \sum_j w_j x_j \quad j: \text{number of features}$$

$$= b + \sum_i c_i g(b_i + \sum_j w_{ij} x_j)$$

i : number of sigmoid functions

- Sigmoid function (S-shaped, Σ -shape)



$$\sigma(b + wx_j) = c \cdot \frac{1}{1 + e^{-(b+wx_j)}}$$

c related to height
 b related to shift
 w related to slope

$$z = \mathbf{w}^T \mathbf{x} + b$$

$$y = \sigma(z) = \frac{1}{1 + e^{-z}} \in (0, 1)$$

Smooth & differentiable

Maps input to (0,1)

Probabilities must lie in the interval (0,1)

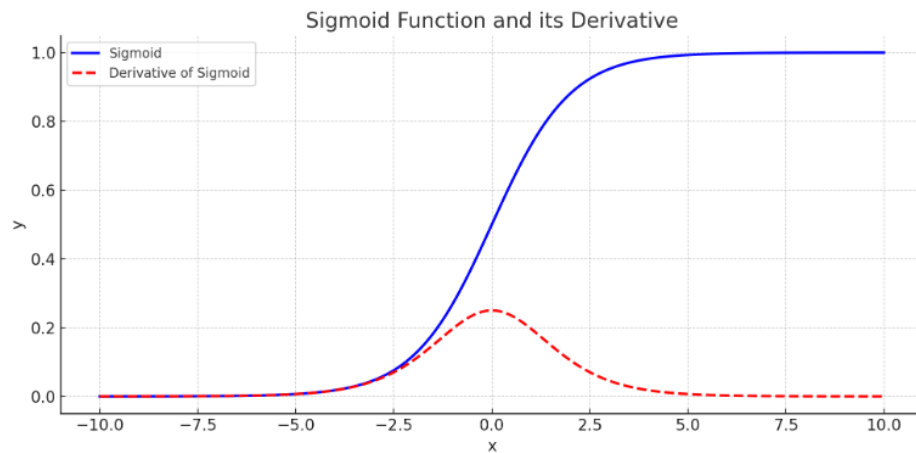


Activation Functions

- Common Activation Functions

Sigmoid Function

 `tf.math.sigmoid(z)`

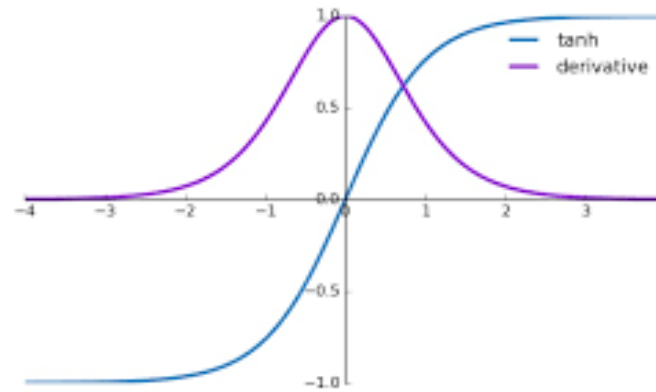


$$g(z) = \frac{1}{1 + e^{-z}}$$

$$g'(z) = g(z)(1 - g(z))$$

Hyperbolic Tangent

 `tf.math.tanh(z)`

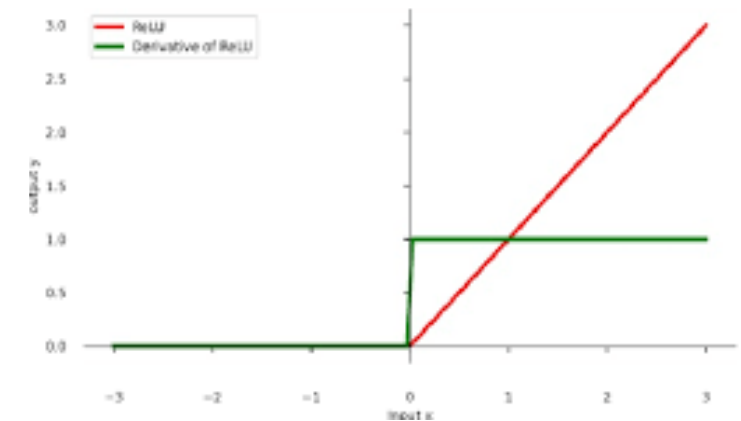


$$g(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

$$g'(z) = 1 - g(z)^2$$

ReLu (Rectified Linear Unit)

 `tf.nn.relu(z)`

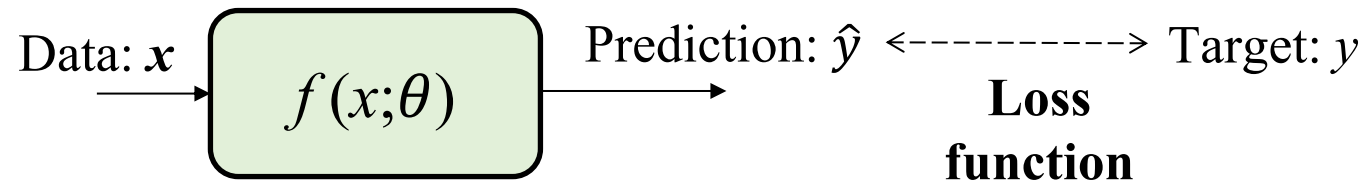


$$g(z) = \max(0, z)$$

$$g'(z) = \begin{cases} 1, & z > 0 \\ 0, & \text{otherwise} \end{cases}$$

A Probabilistic View

- Goal: Given a dataset of input-output pairs (x, y) , our objective is to learn a function $f(x; \theta)$ that maps inputs to outputs.



- From a **probabilistic perspective**, this is equivalent to estimating the most likely output $\hat{y}$ or a given input x under a model parameterized by θ :

discriminative modeling

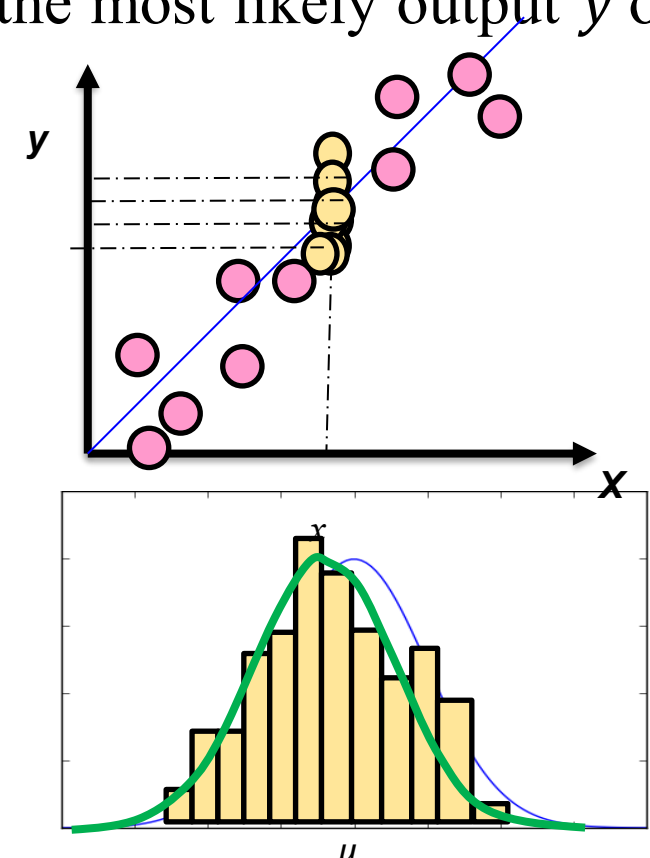
$$\hat{y} = f(x; \theta) = \arg \max_y p(y | x; \theta)$$

- Learning task ---- Regression

Predicting a numerical value based on given input data..

$$f : \mathbb{R}^n \rightarrow \mathbb{R}$$

e.g., temperature forecasting





Maximum likelihood Estimator (MLE)

- Likelihood (似然): quantifies how likely the observed data is, given a set of model parameters θ .
- Goal of MLE: to find the parameter θ that make the observed data most probable:

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} p(\mathcal{D} | \theta)$$

From the god's point of view

$$x \sim G(x | \mu, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

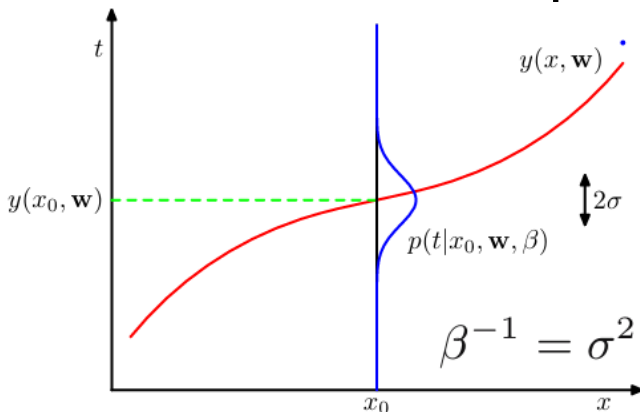
This is the true (but unknown) data-generating distribution.

Sample

Observed Dataset

N samples

$$\mathcal{D} = \{(x_1, y_1), \dots, (x_N, y_N)\}$$



From the human's point of view

$$\hat{\theta} = \arg \max_{\theta} p(\mathcal{D} | \theta) = \arg \max_{\theta} \prod_{n=1}^N p(x_n, y_n | \theta)$$

standardized

$$p(\mathcal{D} | \theta) = p(\mathcal{D} | \mathbf{w}) = \prod_{n=1}^N p(\mathbf{x}_n, y_n | \mathbf{w})$$

Product rule

$$= \prod_{n=1}^N p(y_n | \mathbf{x}_n, \mathbf{w}) p(\mathbf{x}_n | \mathbf{w})$$

irrelevant to $\mathbf{w}$

$$= \prod_{n=1}^N p(y_n | \mathbf{x}_n, \mathbf{w}) p(\mathbf{x}_n)$$

Assuming Gaussian is the distribution

$$= \prod_{n=1}^N G(y_n; \hat{y}_n, \sigma^2) p(x_n) = \prod_{n=1}^N G(y_n; \mu, \sigma^2) p(x_n)$$

$\mathbf{w}^T \mathbf{x}_n$



Maximum likelihood Estimator (MLE)

- Gaussian distribution

$$G(\mathbf{x}|\mu, \sigma^2) = \frac{1}{(2\pi)^{1/2}} \frac{1}{\sigma} \exp \left\{ -\frac{1}{2} \frac{(x - \mu)^2}{\sigma^2} \right\}$$

- Likelihood function

$$p(\mathcal{D}|\mathbf{w}) = \prod_{n=1}^N G(y_n; \mu, \sigma^2) p(x_n) = \prod_{n=1}^N \frac{\beta}{\sqrt{2\pi}} \exp(-\frac{\beta}{2} (y_n - \mathbf{w}^T x_n)^2) P(x_n)$$

- Maximum loglikelihood

- The preference to sums is over products.
- The optimal point does not change since log is monotone increasing.

$$\arg \max_{\mathbf{w}} \log P(\mathcal{D}|\mathbf{w}) =$$

$$\arg \max_{\mathbf{w}} \log \prod_{n=1}^N \sqrt{\frac{\beta}{2\pi}} \exp(-\frac{\beta}{2} (y_n - \mathbf{w}^T \mathbf{x}_n)^2) P(\mathbf{x}_n)$$

$$= \arg \max_{\mathbf{w}} N \log \sqrt{\frac{\beta}{2\pi}} - \frac{\beta}{2} \sum_{n=1}^N (y_n - \mathbf{w}^T \mathbf{x}_n)^2 + \sum_{n=1}^N \log P(\mathbf{x}_n)$$

Ignore terms irrelevant to $\mathbf{w}$

$$\arg \max_{\mathbf{w}} \log P(\mathcal{D}|\mathbf{w})$$

$$= \arg \min_{\mathbf{w}} \frac{\beta}{2} \sum_{n=1}^N (y_n - \mathbf{w}^T \mathbf{x}_n)^2$$

Minimize sum of square error

$$E_D(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (y_n - \mathbf{w}^T x_n)^2$$

$$\mu_{\text{ML}} = \frac{1}{N} \sum_{n=1}^N x_n \quad \sigma_{\text{ML}}^2 = \frac{1}{N} \sum_{n=1}^N (x_n - \mu_{\text{ML}})^2$$

Maximum a Posterior (MAP)

- Problem with MLE:
 1. Relies only on observed data
 2. Can lead to **overfitting**, especially when data is small or noise.
- Purpose of maximum a posterior: Incorporate prior knowledge about parameters $\mathbf{w}$.

$$\hat{\mathbf{w}}_{MAP} = \arg \max_{\mathbf{w}} p(\mathbf{w} | \mathcal{D}) \quad (\text{given } \mathcal{D} \text{ find } \mathbf{w})$$

Combine data (likelihood) with prior belief to estimate parameters.

Product rule of probability

$$P(a, b, c) = P(a|b, c)P(b|c)P(c)$$

Conditional density function

$$P(\mathbf{x} = x | \mathbf{y} = y) = \frac{P(\mathbf{x} = x, \mathbf{y} = y)}{P(\mathbf{y} = y)}, \quad \text{only when } P(\mathbf{y} = y) > 0.$$

- **Probability mass function:**

If $\mathbf{x}$ is discrete, probability mass function $P(\mathbf{x}=x)$ or $P(x)$, ex. the output of a fair dice has discrete uniform distribution with $P(x) = 1/6$.

- **Probability density function:**

If $\mathbf{x}$ is continuous, $P(\mathbf{x} = x)$ denotes a probability density function $p(x)$, rate of increase in probability at x .

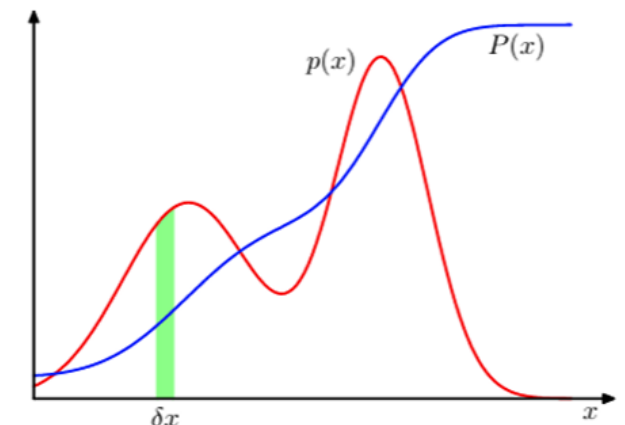


Fig. 1: The probability density $p(x)$ can be expressed as the derivative of a cumulative distribution function $P(x)$



Maximum a Posterior (MAP)

- Bayes' Theorem

$P(\mathbf{y} \mathbf{x}) = \frac{P(\mathbf{x} \mathbf{y})P(\mathbf{y})}{P(\mathbf{x})}$ $posterior = \frac{likelihood \times prior}{evidence}$	e.g.: $\mathbf{x}$: symptoms; $\mathbf{y}$: disease; $P(\mathbf{y})$: the probability of getting this disease $P(\mathbf{x} \mathbf{y})$: probability of this disease having this symptom.
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- Maximum a Posterior (MAP)

Function with variable $\mathbf{w}$

$$p(\mathbf{w} | \mathcal{D}) = \frac{\overset{\text{Likelihood}}{p(\mathcal{D} | \mathbf{w})} \overset{\text{Prior}}{p(\mathbf{w})}}{\underset{\text{Evidence}}{p(\mathcal{D})}}$$

Posterior

Posterior probability $p(\mathbf{w} | \mathcal{D})$: updated belief after seeing data

Likelihood function $p(\mathcal{D} | \mathbf{w})$: fit to data (related to some known parameters based on the observed data)

Prior probability $p(\mathbf{w})$: The probability of $\mathbf{w}$ available before observation of the events.

Evidence $p(\mathcal{D})$: normalization constant (independent of $\mathbf{w}$)

Integrate w.r.t $\mathbf{w} \rightarrow \int p(\mathbf{w}|D)d\mathbf{w} = \int \frac{p(D|\mathbf{w})p(\mathbf{w})}{p(D)} d\mathbf{w} \rightarrow p(D) = \int p(D|\mathbf{w})p(\mathbf{w})d\mathbf{w}$

Total area of posterior probability distribution = 1 $p(D)$ is constant (Not related to $\mathbf{w}$)

Posterior probability $p(\mathbf{w}|D) = \frac{p(D|\mathbf{w})p(\mathbf{w})}{\int p(D|\mathbf{w})p(\mathbf{w})d\mathbf{w}} \rightarrow \text{Posterior} \propto \text{Likelihood} \times \text{Prior}$



Maximum a Posterior (MAP)

- Maximize the possibility of the posterior function to get the most possible parameters $\mathbf{w}$ with the **consideration of prior probability**

$$\arg \max_{\mathbf{w}} P(\mathbf{w}|\mathbf{X}) = \arg \max_{\mathbf{w}} \frac{P(\mathbf{X}|\mathbf{w})P(\mathbf{w})}{P(\mathbf{X})} \quad \text{where } P(\mathbf{w}) \text{ is prior knowledge about } \mathbf{w}.$$

$$= \arg \max_{\mathbf{w}} P(\mathbf{X}|\mathbf{w})P(\mathbf{w})$$

$$= \arg \max_{\mathbf{w}} \log P(\mathbf{X}|\mathbf{w})P(\mathbf{w})$$

$$= \arg \max_{\mathbf{w}} \log P(\mathbf{X}|\mathbf{w}) + \log P(\mathbf{w})$$

Assume $P(\mathbf{w}) \sim G(0, \alpha^{-1}\mathbf{I})$

$$P(\mathbf{w}) = \left(\frac{\alpha}{2\pi}\right)^{\frac{M}{2}} \exp\left(-\frac{\alpha}{2}\mathbf{w}^T\mathbf{w}\right)$$

Then,

$$\arg \max_{\mathbf{w}} P(\mathbf{w}|\mathbf{X}) =$$

$$= \arg \min_{\mathbf{w}} \frac{\beta}{2} \sum_{n=1}^N (y_n - \mathbf{w}^T \mathbf{x}_n)^2 + \frac{\alpha}{2} \mathbf{w}^T \mathbf{w}$$

$$= \arg \min_{\mathbf{w}} \sum_{n=1}^N (y_n - \mathbf{w}^T \mathbf{x}_n)^2 + \frac{\lambda}{2} \mathbf{w}^T \mathbf{w}$$

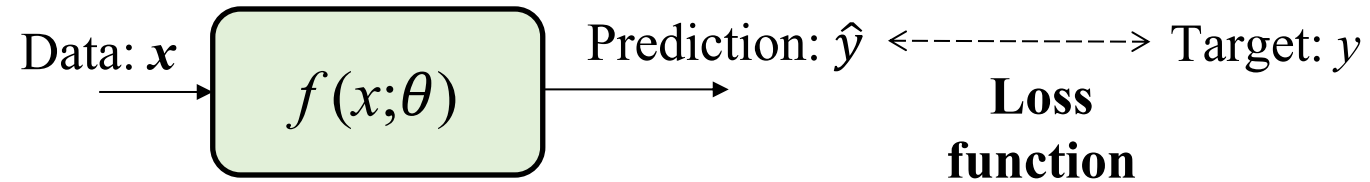
If $p(w)$ is uniform (constant), MAP reduces to MLE.

If $p(w)$ is Gaussian, MAP corresponds to MLE with L2 regularization (weight decay).



Supervised Learning Tasks

- Supervised Learning Tasks: The performance at task improves with experience.



- Regression: predict a numerical value given some input

$$f : \mathbb{R}^n \rightarrow \mathbb{R}^m \text{ e.g., temperature forecasting}$$

- Classification: predict a categorical label given some input

$$f : \mathbb{R}^n \rightarrow \{1, \dots, k\} \text{ e.g., classifying machine parts as defective or non-defective}$$